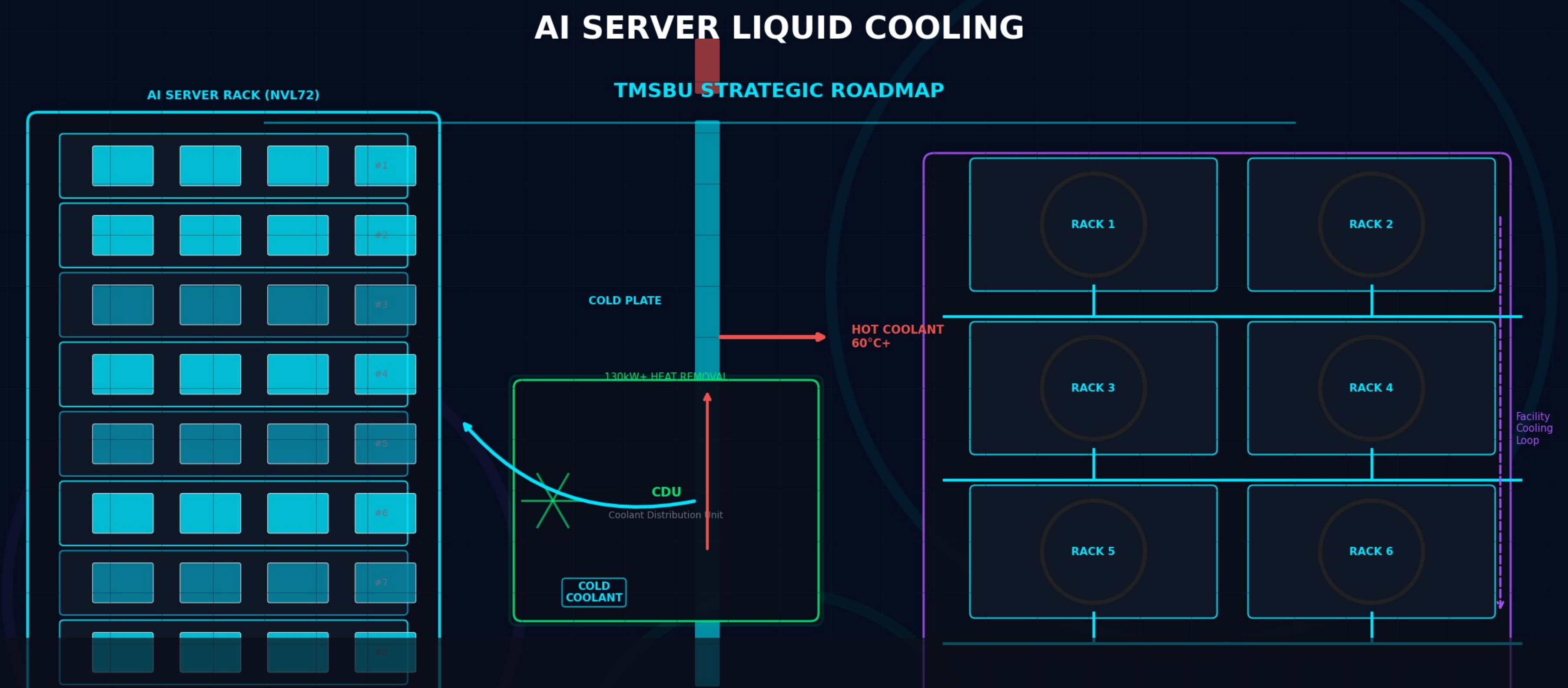


AI Server Liquid Cooling:

TMSBU Commercial Opportunity & Strategic Roadmap

From Component Supplier to Thermal System Solution Provider

Executive Summary

液冷不是產品升級，而是商業模式的轉型

Liquid cooling is not only a product opportunity — it is a business model transition for TMSBU

01 TREND

AI GPU / Rack Power Exceeds Air Cooling

GPU TDP 突破 700W ， GB200 NVL72 rack power 達 130kW+ 。風冷已達物理瓶頸，散熱成為 AI 基礎建設擴張的硬約束。

02 MARKET

Liquid Cooling Adoption Accelerating

液冷滲透率： 2023 年 20% → 2026 年 60% → 2030 年 80%+ 。從選配變成新建 AI 資料中心的標準架構。

03 HYBRID

Hybrid Air + Liquid Remains Necessary

GPU 本體用液冷，但 memory 、 PCB 、 power delivery 、 networking 仍需風冷。兩種散熱方式長期共存， TMSBU 兩者皆受益。

04 OPPORTUNITY

TMSBU Can Expand into Module & System

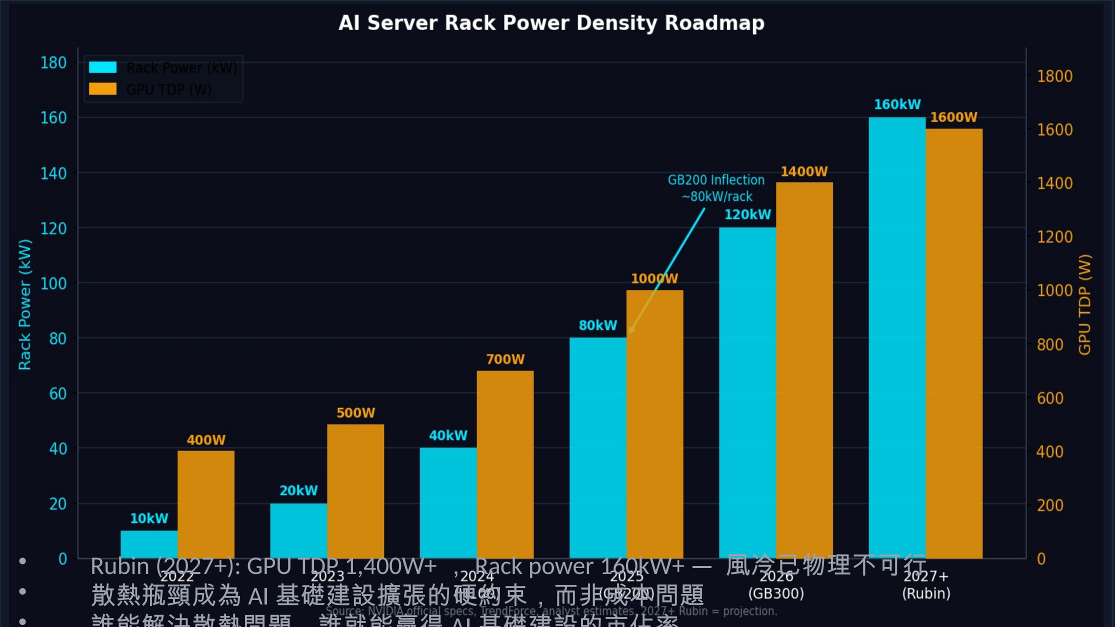
Delta TMSBU 已有 Fan 、 HS 、 HP 、 VC 、 Cold Plate 製造基礎。從零件供應商升級為模組與系統層級熱管理方案供應商。

05 ACTION

6–18 Months Are Critical

Market Inflection: AI Compute Density Creates Thermal Bottleneck

AI compute density is growing faster than traditional thermal architecture can support



- Rubin (2027+): GPU TDP 1,400W+ ，Rack power 160kW+ — 風冷已物理不可行
- 散熱瓶頸成為 AI 基礎建設擴張的硬約束，而非成本問題
- 誰能解決散熱問題，誰就能贏得 AI 基礎建設的市佔率

GPU TDP — **400W → 1,400W**

Per chip, 2022–2027+

Rack Power — **10kW → 160kW**

Per rack, 2022–2027+

Inflection — **GB200 NVL72**

~80–130kW/rack, liquid required

Market — **$4.2B → $39.1B**

Liquid cooling TAM by 2030 (M&M)

Why Air Cooling Alone Is No Longer Enough

Not air vs. liquid — but hybrid air + liquid is the new standard architecture

Air Cooling Still Required

- Memory modules (HBM)
- PCB / power delivery
- Storage / networking
- Low-power accessories

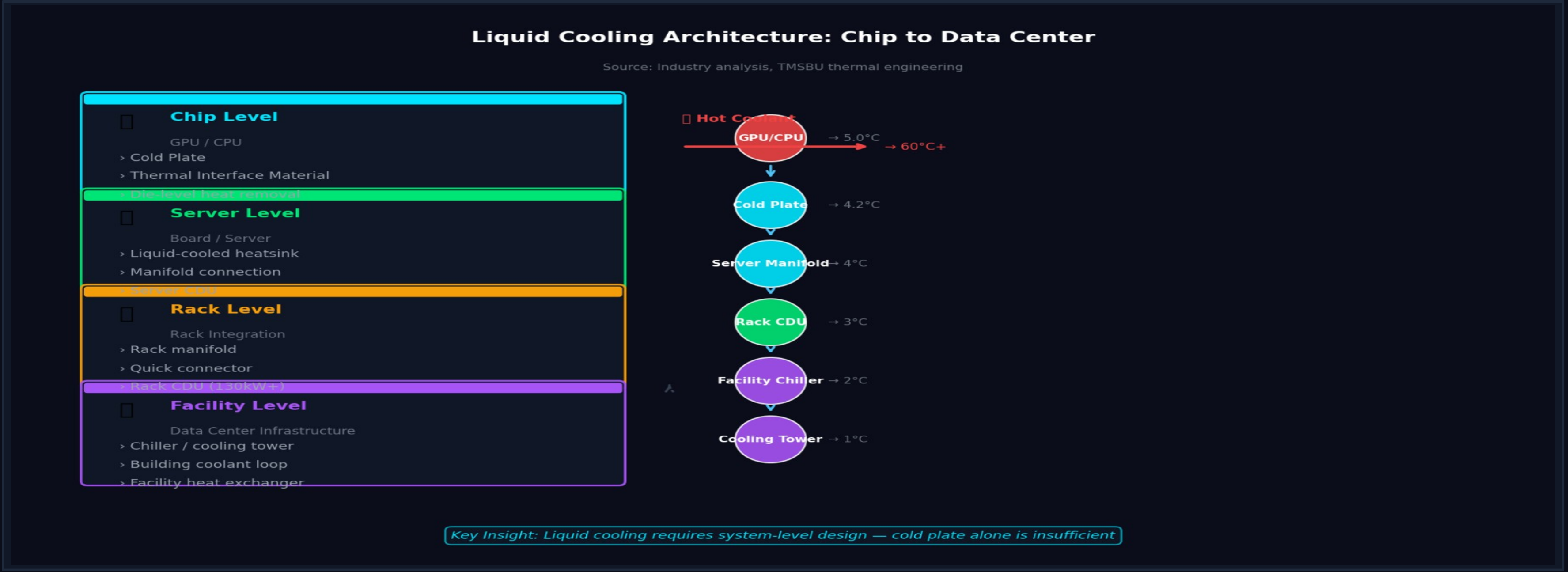
Air Cooling Limitations

- Heat flux 上限 ~50W/cm²， GPU 已超標
- Fan 噪音與功耗隨密度非線性增加
- Rack power > 30kW 時， TCO 急速上升
- PUE penalty ：高密度風冷機房 PUE > 1.4

KEY INSIGHT: Hybrid = Liquid handles GPU/CPU hotspot + Air handles memory/PCB/power. TMSBU benefits from BOTH trends.

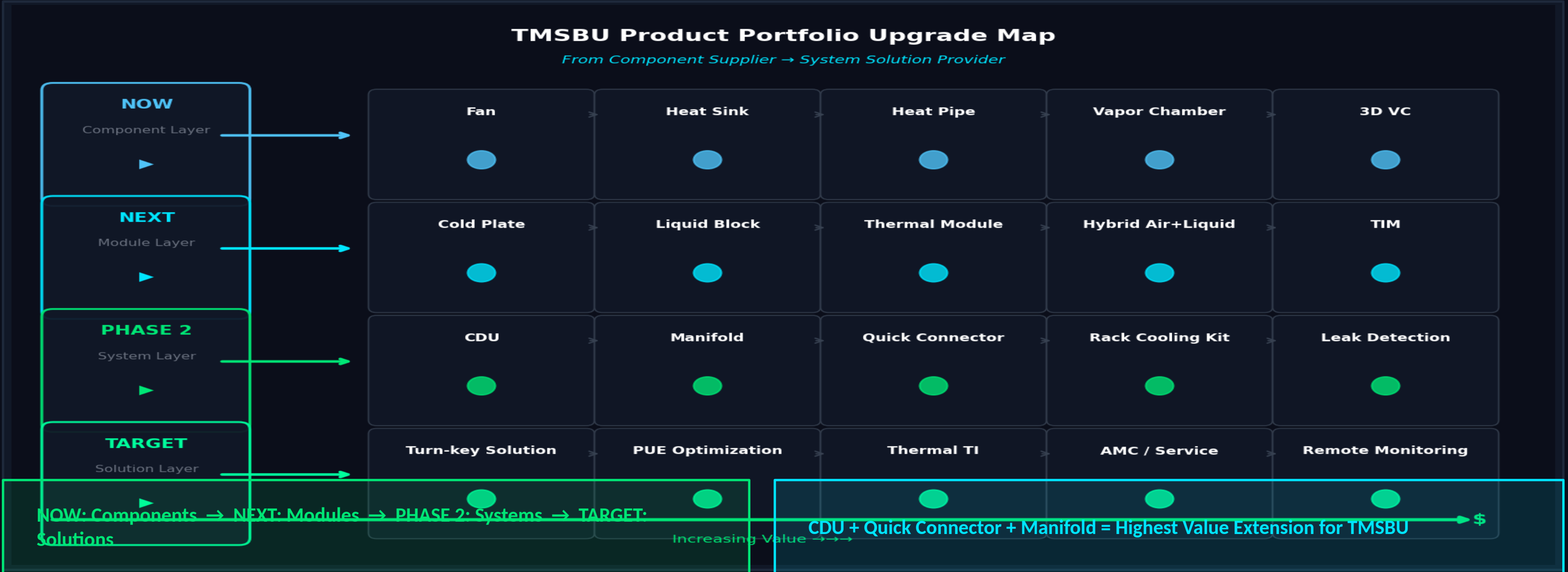
Liquid Cooling Architecture Becomes the New Standard

液冷的價值不在冷板，在於完整系統整合： Cold Plate + CDU + Manifold + Quick Connector + Leak Detection



TMSBU Portfolio Mapping: From Air Cooling to Liquid Cooling

Delta TMSBU has a natural expansion path from existing thermal components into liquid cooling modules and systems



NOW: Components → NEXT: Modules → PHASE 2: Systems → TARGET: Solutions

CDU + Quick Connector + Manifold = Highest Value Extension for TMSBU

TAM / SAM / SOM: Market Opportunity for TMSBU

The addressable opportunity is larger than cold plates alone — system-level cooling multiplies TMSBU's reachable market

TAM
Total AI Data Center Liquid Cooling — **$39.1B** — by 2030

Cold Plate
Chip-level direct cooling — **$5.2B** — ~13% of TAM

CDU
Coolant distribution + heat rejection — **$3.8B** — ~10% of TAM

Manifold + QC
Rack distribution + connectors — **$3.8B** — ~10% of TAM

Hybrid Air
Fan + HS + VC for non-GPU components — **$4.5B** — Additional market

SOM — **$1.8B** — Assumption-based
TMSBU realistic addressable (×14%)

Competitive Positioning & Differentiation

Avoid competing as a low-margin component supplier — move toward differentiated thermal system partnership

Traditional Thermal Suppliers Nidec, Sunon, Global Sunon	+ Fans, HS, VC	− No liquid cooling system
Liquid Cooling Specialists CoolIT, Rittal, Avery Dennison	+ CDU, cold plate, manifold	− Limited air cooling synergy
Server ODMs (In-house) Foxconn, Wistron, Inventec	+ Full system integration	− Outsource thermal to TMSBU
Connector / CDU Specialists Staubli, Liiver, CPC	+ Quick connector, CDU	− Narrow component focus
Integrated DC Infra Schneider, Vertiv, Rittal	+ Full data center solutions	− Not chip-level thermal

TMSBU Differentiators

Air + Liquid Hybrid
唯一同時具備兩種散熱技術的供應商

Delta Group Synergy
Fan + Power + Control + DC Infra

Manufacturing Scale
現有散熱零件製造基地，快速延伸

Chip to Rack Capability
VC/HP → Cold Plate → Rack CDU 全段覆蓋

Customer Relationships
現有服務服務器 ODM/OEM 的客戶基礎

Capability Gap & Risk Assessment

The opportunity is attractive, but TMSBU must close key capability gaps before scaling

Risk Item	Impact	Urgency	Key Action
CDU Design & Manufacturing	HIGH	HIGH	Core system component. Partner or acquire.
Quick Connector Qualification	HIGH	MED	Reliability is critical. Staubli/Liiver partnership.
Leak Detection & Reliability	HIGH	HIGH	CDU failure = server damage. Must be bulletproof.
Coolant Compatibility	MED	MED	OEM-specific coolant specs. Early engagement needed.
System-Level Validation	HIGH	HIGH	Need rack-level thermal testing capability.
Thermal Simulation Tools	MED	MED	Chip-to-rack multi-physics simulation required.
Customer Qualification Cycle	MED	HIGH	12-18 month design-in cycles. Start NOW.
Price Competition (China)	MED	MED	Cost-down pressure on cold plates. Differentiate on quality.
Integration Responsibility	HIGH	MED	Warranty boundary unclear in system selling.
Global Support Network	MED	LOW	Needed for Tier-1 hyperscaler contracts.

BOTTOM LINE: Highest priority risks are CDU capability, leak reliability, and customer qualification cycle. All three must be addressed in Phase 1 (0-3 months).

Strategic Roadmap & Recommended Actions (6–18 Months)

Execute a phased roadmap: Component expansion → Module development → System integration → Solution business

Phase 1 0–3 mo	Phase 2 3–6 mo	Phase 3 6–12 mo	Phase 4 12–18 mo